2 Force and Laws of Motion

"We can control the magnitude of the applied force and also its direction, so force is a vector quantity, just like velocity and acceleration."

A cricketer strikes the ball with his bat, applying a force and directing the ball into a particular part of the ground. He can hit the ball at different speeds and direct it into different parts of the ground.

1. Introduction

In common usage, a force is a push or a pull on some object. If you pull hard enough on a trolley [see figure (a)], the trolley moves. If a player kicks the football [see figure (b)], the football moves.

(a) A trolley moves when it is pulled

(b) Player kicking a football

Examples of some contact forces

☞ There are two types of forces : contact forces and non-contact forces. The types of contact forces are muscular force, tension, friction etc. The types of non-contact force are electrostatic force, magnetic force, gravitational force.

Do You Remember ?

★ Frictional force not only opposes motion but also helps a body to move. It is friction that helps us walk and run.

★ The earth is acted upon by the gravitational pull of the stars, planets, sun and moon. However, the gravitational pull of the sun is the most dominant.

2. Effects of force

(1) A force can distort an object i.e. it can change the shape and size of an object.

(2) A force can start an object at rest i.e. it can move a stationary object.

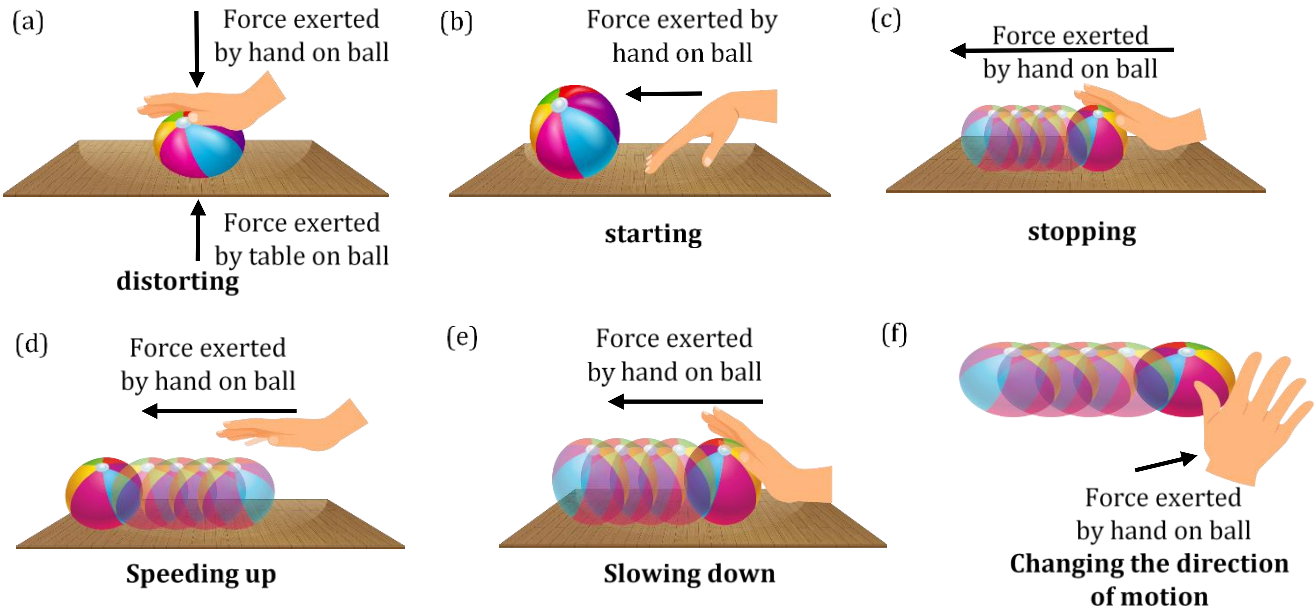
(3) A force can stop a moving object i.e. it can cause a moving object to come to rest.

(4) A force can change the speed or the magnitude of velocity of an object i.e. it can increase or decrease the speed of an object.

(5) A force can change the direction of a moving object.

Petronas Twin Towers in Kuala Lumpur, Malaysia, the world's tallest twin towers. The design of tall buildings involves understanding forces. Towering buildings are susceptible to movement from the wind.

☞ A **force** can be defined as 'a push or a pull exerted on an object that can cause the object to speed up, slow down, or change direction as it moves or it can change its shape and size'.

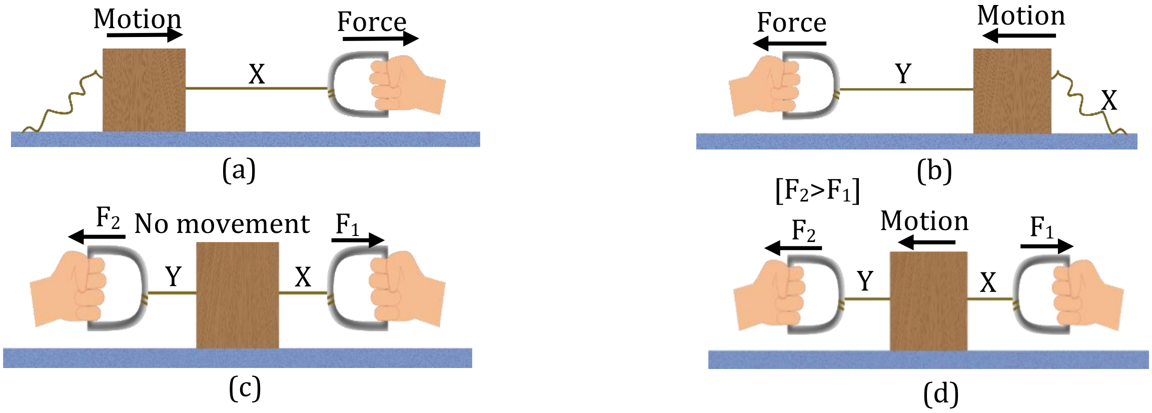


Effects of force

Active Physics 1

1. Figure shows a wooden block on a horizontal table. Two strings X and Y are tied to the two opposite faces of the block as shown. Apply a force by pulling the string X, the block begins to move to the right [see figure (a)]. Similarly, if we pull the string Y, the block moves to the left [see figure (b)].

2. Now, pull both strings X and Y simultaneously i.e. in opposite direction. If the forces on the block are equal, the block will not move [see figure (c)]. If the opposite forces on the block are of different magnitudes, the block will move in the direction of the greater force [see figure (d)].



Active physics 1

Conclusion : When a single force is applied on an object, the object will move in the direction of applied force. When two or more forces are applied on an object, the object will move in the direction of net force acting on the object. Thus, **force is a vector quantity**.

> The equilibrium rule : For any object or system of objects in equilibrium, the sum of the forces acting on it equals zero.
>
> SPOT LIGHT

3. Balanced and unbalanced forces

Balanced forces

If the resultant of all forces acting on a body is zero, the forces are called **'balanced forces'**.

☞ If the net force exerted on an object is zero, then the forces acting on it are said to be **balanced**. In such a case, the acceleration of the object is zero and its velocity remains constant. That is, if the net force acting on the object is zero, the object either remains at rest or continues to move with constant velocity.

Unbalanced forces

If the resultant of all forces acting on a body is not zero, the forces are called '**unbalanced forces'**.

- If the net force exerted on an object is not zero, then the forces acting on it are said to be **unbalanced**. In such a case, the acceleration of the object is not zero and its velocity changes. That is, if the net force acting on the object is not zero, then such a force changes state of rest or the state of uniform motion of the object.

  If there is an unbalanced force acting on an object, the change in its velocity would continue as long as this unbalanced force persists. If this force is removed completely, the object would continue to move with the velocity it has acquired till then.
- An object is in equilibrium when it has zero acceleration. This means, it is the state of an object or system of objects for which there are no changes in motion. It includes the state of rest as well as the state of uniform motion.

Quick Tips

★ If $F_{net} = 0$, forces are balanced.

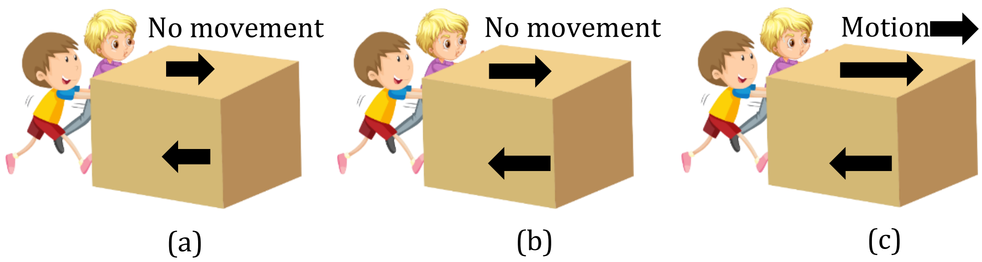
★ If $F_{net} \neq 0$, forces are unbalanced.

Building Concepts 1

What happens when some children try to push a box on a rough floor?

Explanation

If they push the box with a small force, the box does not move because of friction acting in a direction opposite to the push [see figure (a)]. This frictional force arises between two surfaces in contact. Here, the friction is between the bottom of the box and floor's rough surface. It balances the pushing force and therefore the box does not move. If the children push the box harder [see figure (b)], the box still does not move. This is because the frictional force still balances the pushing force. If the children continue to increase the push force on the box, at some point the pushing force becomes bigger than the frictional force [see figure (c)]. That is, there is an unbalanced force due to which the box starts moving (accelerating).



Building Concepts 1

Check your Concepts 1

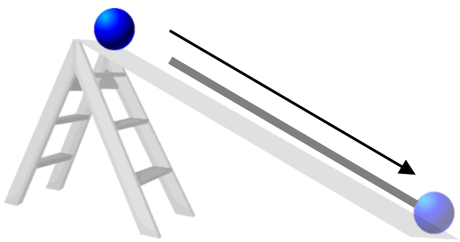
1. What happens if we stop pedaling while riding a bicycle?

2. Is it correct to say that 'an object maintains its motion under the continuous application of an unbalanced force'?

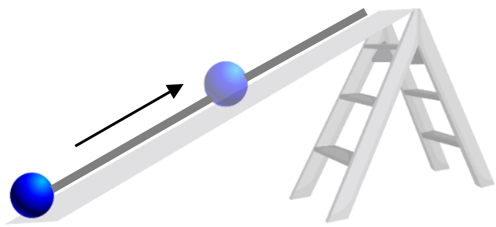
3. Which type of force is required to accelerate or retard the motion of an object?

4. Galileo's inclined planes

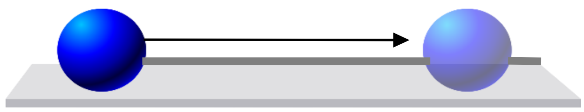
Galileo studied motion of objects on an inclined plane. He noted that balls rolling down [see figure (a)] the inclined planes picked up speed (i.e. acceleration), while balls rolling on [see figure (b)] up the inclined planes lost speed (i.e. retardation). From this he reasoned that balls rolling on [see figure (c)] a horizontal plane would neither speed up nor slow down. The ball would finally come to rest not because of its 'nature' but because of friction. This idea was supported by Galileo's observation of motion along smoother surfaces. When there was less friction, the motion of objects persisted for a longer time. The smaller the friction, the more the motion approached constant velocity. Galileo concluded that an object moving on a frictionless horizontal plane must neither have acceleration nor retardation, i.e. it should move with constant velocity.



(a) A ball rolling down the plane. Velocity of ball increases (accelerated motion).



(b) A ball rolling up the plane. Velocity of ball decreases (retarded motion).



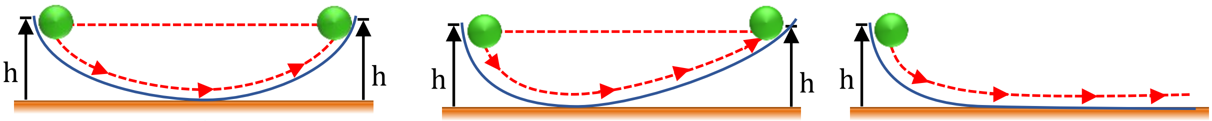
(c) A ball rolling on a horizontal plane. Velocity of ball remains constant (uniform motion).

Motion of balls on different planes

Another experiment by Galileo leading to the same conclusion involves a **double inclined plane**. A ball released from rest on one of the planes rolls down and climbs up the other. If the planes are smooth, the final height of the ball is nearly the same as the initial height (a little less but never greater). In the ideal situation, when friction is absent, the final height of the ball is the same as its initial height [see figure (a)]. If the slope of the second plane is decreased and the experiment is repeated, the ball will still reach the same height, but in doing so, it will travel a longer distance.

Galileo concluded that 'a ball rolling down the first inclined plane on the left tends to roll up to its initial height on the second plane on the right, thus, the ball must roll a greater distance as the angle of the second inclined plane on the right is reduced' [(see figure (b)]. He argued that when the slope of the second plane is made zero i.e. it becomes a horizontal plane, the ball must travel an infinite distance since it can never reach its initial height on first plane [(see figure (c)]. In other words, its motion never ceases. This is, of course, an idealised situation.

☞ Galileo arrived at a new insight that 'the state of rest and the state of uniform motion (motion with constant velocity) are equivalent'. In both cases, there is no net force acting on the body. It is incorrect to assume that a net force is needed to keep a body in uniform motion. To maintain a body in uniform motion, we need to apply an external force to counter the frictional force, so that the two forces sum up to zero net external force. A body does not change its state of rest or uniform motion, unless an external force compels it to change that state. The tendency of things to resist changes in motion was what Galileo called **inertia**.



(a) (b) (c)

Galileo's observations on motion of a ball on a double inclined plane

5. Newton's first law of motion

Newton built on Galileo's ideas and laid the foundation of mechanics in terms of three laws of motion that go by his name. Galileo's **law of inertia** was his starting point which he formulated as the **first law of motion :**

'Every object continues in its state of rest, or of uniform motion in a straight line, unless it is compelled to change that state by forces impressed upon it'.

Concept of Inertia

Inertia is 'the natural tendency of an object to remain at rest or in motion at a constant speed along a straight line'. In other words, 'the tendency of an object to resist any attempt to change its velocity' is called **inertia**.

Be Alert !

★ It should be noted that Newton's first law of motion is also called **Galileo's law of inertia**.

Building Concepts 2

Do all bodies have same inertia? Is there any relation between inertia and mass of body?

Explanation

We know that it is easier to push an empty box than a box full of books. Similarly, if we kick a football it flies away. But if we kick a stone of the same size with equal force, it hardly moves. We may, in fact, get an injury in our foot while doing so A force that is just enough to cause a small cart to pick up a large velocity will produce a negligible change in the motion of a train. This is because, in comparison to the cart the train has a much lesser tendency to change its state of motion. Accordingly, we say that the train has more inertia than the cart.

Clearly, heavier or more massive objects offer larger inertia. Quantitatively, the inertia of an object is measured by its mass.

☞ The **mass** of an object is a quantitative measure of **inertia**. More the mass, more will be the inertia of an object and vice-versa.

Inertia of an object can be of three types :

(1) Inertia of rest (2) Inertia of motion (3) Inertia of direction

Inertia of rest

It is the tendency of an object to remain at rest. This means an object at rest remains at rest until a sufficiently large external force is applied on it.

Examples of inertia of rest

(1) When you are sitting in a stationary car, if the car starts suddenly i.e. accelerates forward, you feel as if your body is being pushed back against the seat, because your body which was initially at rest resists this change due to inertia. The lower part of body comes in motion as it is in direct contact with the car floor, while the upper portion still remains at rest due to inertia of rest [see figure]. If the speed of car increases slowly, you will not feel a push or a jerk because the inertia of motion will get transferred to the whole body.

Sudden start of a car

Example of inertia of rest

(2) When a blanket is given a sudden jerk, the dust particles fall off. This is because the blanket suddenly comes in motion but the dust particles still remain at rest. As a result, the dust particles get separated from the blanket.

Inertia of motion

It is the tendency of an object to remain in the state of uniform motion. This means an object in uniform motion continues to move uniformly until an external force is applied on it.

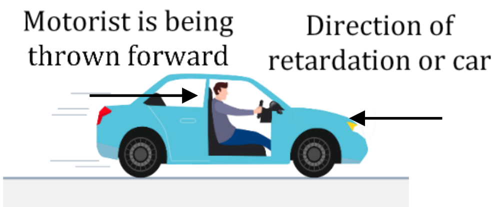
Examples of inertia of motion

(1) When you are driving a car and you apply brakes to stop the car suddenly, you feel as if your body is being pushed forward, because your body resists the decrease in speed. The lower part of body comes to rest as it is in direct contact with the car floor, while the upper portion still remains in motion due to inertia of motion [see figure(a)]. If you stop the car by decreasing its speed slowly, you will not feel a push or a jerk because the inertia of rest will get transferred to the whole body.

(2) A person jumping out of a moving train has the tendency to fall forward. This is because on jumping, his feet come to rest as they touch the ground. But, his upper body continues to move forward due to inertia of motion.

(3) An athlete runs for some distance quickly before taking a long jump. As a result, he takes a longer jump due to inertia of motion [see figure (b)].

(4) When you move a hammer with loose hammerhead in downward motion and suddenly stop it on a floor or a wooden base, the hammerhead gets tightened. This is because the handle of the hammer suddenly comes to rest on hitting the floor, while the hammerhead continues to move downward due to more inertia of motion, and hence gets tightened [see figure (c)]. If you move the hammer slowly, the state of rest will get transferred to the hammerhead also, thus, the hammerhead will not get tightened.



Sudden stop of a car

(a)


Inertia of motion
Helps a long jumper

(b)


(c) Tightening of hammerhead

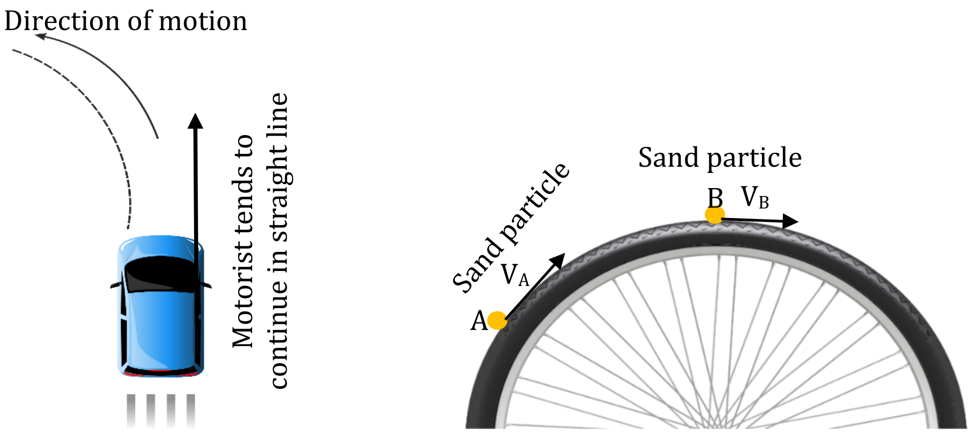
Examples of inertia of motion

Inertia of direction

It is the tendency of an object to maintain its direction. This means an object moving in a particular direction continues to move in that direction until an external force is applied to change it.

Examples of inertia of direction

(1) When your motorcar makes a sharp turn at a high speed, you tend to get thrown to one side. You tend to continue in straight-line motion due to inertia of direction [see figure (a)].



(a) (b)

Examples of inertia of direction

(2) When a wheel rotates at high speeds, the sand particles on the wheel fly tangentially along a straight line due to inertia of direction [see figure (b)].

Check your Answers 1

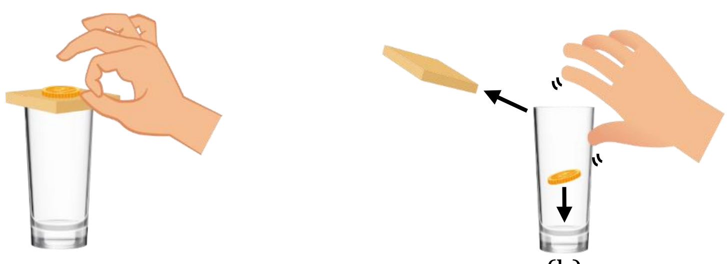
1. When we stop pedaling, the bicycle begins to slow down. This is because of the frictional force acting opposite to the direction of motion. In order to keep the bicycle moving, we have to start pedaling again.

2. Yes, it is incorrect. An object moves with a uniform velocity when the forces (e.g. pushing force and frictional force) acting on the object are balanced i.e. there is no net external force acting on it. If an unbalanced force is applied on the object, there will be a change either in its speed or in the direction of its motion.

3. To accelerate or retard the motion of an object, an unbalanced force is required. The change in its speed (or in the direction of motion) will continue as long as this unbalanced force is applied.

Active Physics 2

1. Set a five-rupee coin on a stiff smooth card covering an empty glass tumbler standing on a table (see figure). Give the card a sharp horizontal flick with a finger.



(a) (b)

Active physics 2

2. If we do it fast then the card moves away, allowing the coin to fall vertically into the glass tumbler due to its inertia of rest. The inertia of the coin tries to maintain its state of rest even when the card flies off.

Active Physics 3

1. Make a pile of similar carrom coins on a table (see figure). Attempt a strong horizontal hit at the bottom of the pile using another carrom coin or the striker. If the hit is strong enough, the bottom coin moves out quickly. Once the lowest coin is removed, the inertia of the other coins makes them 'fall' vertically on the table.

2. This is because the lowest coin comes in motion while the other coins remain at rest due to inertia. If the hit is not so strong, the inertia of motion is transferred to all the coins, thus, the coins may fall randomly with or without the actual movement of lowest coin.

Active physics 3

☞ **Air bags in car- a safety device:** If a head-on collision of a car is violent enough, sodium azide undergoes a rapid chemical reaction to produce non-toxic nitrogen gas, which inflates an airbag. The inflated airbag provides a protective cushion to slow down the head and body of a motorist.

(a) (b) (c)

Airbag systems in cars are designed to safeguard the motorist during vehicle collisions

Check your Concepts 2

1. Suppose you are standing in a bus which is at rest. What happens when the bus starts suddenly?

2. What happens when you shake vigorously a branch of a mango tree?

3. Why a bullet shot on a tightly fitted window glass sheet makes a small circular hole, while a cricket ball damages a big portion of the same sheet?

6. Linear momentum (or momentum)

The linear momentum of a particle or an object that has a mass 'm' moving with a velocity 'v' is defined to be the product of the mass and velocity.

☞ Linear momentum is a **vector quantity,** because it equals the product of a scalar quantity 'm' and a vector quantity 'v'. The direction of linear momentum is 'the direction along the velocity'.

$$\mathbf{p = m \times v}$$

☞ If there is an unbalanced force acting on an object, its velocity changes, hence, its momentum also changes. If the forces acting on an object are balanced, its velocity is constant, hence, its momentum is also constant.

☞ The linear momentum of a particle is

(i) directly proportional to its mass (ii) directly proportional to its velocity

Units of linear momentum

SI unit : kg m/s or kg m s^{-1} (Another SI unit of momentum is Newton-second or N-s)

C.G.S. unit : g cm/s or g cm s^{-1}

Quick Tips

★ Every moving body possesses momentum.

★ Momentum is the combined effect of mass and velocity.

★ Momentum is considered to be a measure of the quantity of motion of a moving body.

★ Force can change the velocity of an object. Thus, force can change the momentum of an object.

★ When an object is moving along a circular path, its velocity is tangential to the circular path, hence, its momentum is also tangential to the circular path.

★ In Newton's second law, F = 0 implies a = 0. Thus, Newton's second law is obviously consistent with Newton's first law.

Building Concepts 3

(i) A car and a truck both are moving with same velocities. Which one has more momentum?

(ii) A car and a truck have same momentum. Which one has more velocity?

Explanation

(i) For a given velocity, the momentum is directly proportional to the mass of the object. This means more the mass, more will be the momentum and vice-versa. Here, velocity of both truck and car is same. Since, the mass of a truck is greater than the mass of a car, therefore, the momentum of truck is more than the momentum of car.

(ii) For a given momentum, the velocity is inversely proportional to the mass of the object. This means smaller the mass, more will be the velocity of an object and vice-versa. Here, momentum of both truck and car is same. Since, the mass of a car is smaller than the mass of a truck, therefore, the velocity of car is more than the velocity of truck.

Numerical Ability 1

1. Momentum of an object is 20 kg m s^{-1}. What will be its momentum if

(a) its mass is doubled but the velocity remains the same?

(b) the velocity is reduced to (1/3) of its original magnitude but mass remains the same?

Solution

Let the mass of the object be m and its velocity be v.

Initial value of momentum, $p = mv = 20$ kg m s^{-1}

(a) New mass, $m' = 2m$, velocity remains the same i.e. $v' = v$

New value of momentum, $p' = m'v' = (2m)(v)$

$= 2mv = 2 \times 20 =$ **40 kg m s^{-1}**

(b) New velocity, $v' = v/3$, mass remains the same i.e., $m' = m$

New value of momentum, $p' = m'v' = m(v/3) = (mv)/3$

$= 20/3 =$ **6.67 kg m s^{-1}**

Decode the problem

To find the momentum, we should know the mass and velocity of the object.

Apply the formula

$p = mv$

2. Calculate the momentum of a motor car of mass 800 kg moving with a velocity of 10 m/s.

Solution

Given,

mass = 800 kg; velocity,

v = 10 m/s

$p = mv = 800 \times 10$

= **8000 kg m/s**

Decode the problem

To find the momentum, we should know the mass and velocity of the object.

Apply the formula

$p = mv$

3. A 65-kg girl is driving a 535-kg car at a constant velocity of 11.5 m/s. Calculate the momentum of the girl-car system.

Solution

Since, we have to find the momentum of the girl-car system, the total mass of the system,

m = mass of girl + mass of car = 65 + 535 = 600 kg

Velocity of car, v = 11.5 m/s

Now, momentum,

$p = mv = 600 \times 11.5$

= **6900 kg m s^{-1}**

Decode the problem

To find the momentum, we should know the mass and velocity of the object.

Apply the formula

$p = mv$

4. The momentum of a 75-g bullet is 9 kg m/s. What is the velocity of the bullet?

Solution

Given, mass of bullet, m = 75 g = (75/1000) kg = 0.075 kg

momentum, p = 9 kg m/s ; velocity, v =?

Now, momentum, p = mv

or $v = \frac{p}{m} = \frac{9}{0.075} = \frac{9 \times 1000}{75}$

= **120 m/s**

Decode the problem

To find the velocity we should know the momentum and mass of the object.

Apply the formula

p = mv

7. Newton's second law of motion

According to the second law of motion, **'the rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts'**.

Mathematically, the force, $F \propto \frac{\Delta p}{t}$

where, Δp = change in momentum, t = time interval.

Mathematical formulation of second law of motion

Let an object of mass 'm' is moving along a straight line with an initial velocity 'u'. It is uniformly accelerated to velocity 'v' in time 't' by the application of a constant force F throughout the time t. The initial and final momentum of the object will be, p_1 = mu and p_2 = mv, respectively.

The change in momentum, $\Delta p = p_2 - p_1 = mv - mu = m(v - u)$

The force F is proportional to the rate of change of momentum, that is,

$F \propto \frac{\Delta p}{t}$ or $F \propto \frac{m(v-u)}{t}$

or $F = k\frac{m(v-u)}{t}$ ...(1)

where, k is a constant of proportionality.

Now, acceleration, $a = \frac{(v-u)}{t}$ ...(2)

From eq. (1) & eq. (2), we get, F = k m a ...(3)

The SI units of mass and acceleration are kg and m s^{-2} respectively. The unit of force is so chosen that the value of the constant 'k' becomes one.

That is, 1 unit of force = k × (1 kg) × (1 m s^{-2}) or k = 1

Thus, the value of k becomes 1. Therefore, the eq.(3) reduces to,

$\boxed{F = ma}$

Unit of force

SI unit : Newton (N)

Since, $F = ma$

or $1\ N = 1\ kg \times 1\ m\ s^{-2} = 1\ kg\ m\ s^{-2}$

$\therefore$ **C.G.S unit :** Dyne

1 dyne $= 1\ g\ cm\ s^{-2}$

$1\ N = 1000\ g \times 100\ cm\ s^{-2} = 10^5\ g\ cm\ s^{-2}$,

or **1 N = $\mathbf{10^5}$ dynes**

Newton's second law can also be stated as : 'the force acting on an object is directly proportional to its mass and the acceleration produced in it'

SPOT LIGHT

Quick Tips

★ Force is not always in the direction of motion. It may be along velocity, opposite to velocity or make angle with velocity.

★ In every motion, force is parallel to acceleration.

★ No force is required to move a body with a uniform velocity.

$\boxed{a = 0, F = 0}$

★ The cause of every accelerated motion is an external force.

★ $a = \dfrac{F}{m}$, the acceleration produced in a body is directly proportional to the force acting on it and inversely proportional to the mass of the body.

Be Alert !

★ If the mass of a body is doubled, its acceleration will be halved and if the mass is halved than acceleration will get doubled (provided the force remains the same).

★ a = –ve, shows that the acceleration is acting in a direction opposite to the motion of the body.

★ F = –ve, shows that force is acting in a direction opposite to that in which the body is moving.

Check your Answers 2

1. When the bus starts suddenly, we feel a push in backward direction. This is because the lower part of our body which is in direct contact with the bus floor starts moving while the upper part is still at rest due to inertia.

2. When we shake vigorously a branch of a mango tree, the mangoes fall down. This is because on shaking, the branches comes in motion while the mangoes still remain at rest due to inertia. Thus, the mangoes get detached and fall.

3. The bullet strikes the glass sheet with a very high speed. Thus, only the portion where the bullet strikes comes in motion while the remaining portion of glass sheet still remains at rest due to inertia. The cricket ball strikes the glass sheet with relatively lower speed. Thus, inertia of motion gets transferred to a big portion of the glass sheet, damaging a big portion of glass sheet.

☞ If force is constant i.e., F = ma = constant, then, the acceleration produced in the body, $a \propto \frac{1}{m}$. That is, for a given force, acceleration produced is inversely proportional to its mass.

☞ If same force F is applied to masses m_1 and m_2 and the resulting accelerations in them are a_1 and a_2 respectively, then,

$m_1a_1 = m_2a_2$ or $\boxed{\frac{a_2}{a_1} = \frac{m_1}{m_2}}$

Building Concepts 4

Equal forces are applied to a car and a truck which produce acceleration in both of them. Which one has smaller acceleration?

Explanation

For a given force, the acceleration produced in the object is inversely proportional to the mass of the object ($a \propto 1/m$). This means more the mass of an object, less will be the acceleration produced in it. Here, force applied on the truck as well as the car is same. Since, the mass of a truck is greater than the mass of a car, therefore, the acceleration of the truck is smaller than the acceleration of the car.

Applications/results of Newton's second law

(i) Suppose a light-weight vehicle (say a small car) and a heavy-weight vehicle (say a loaded truck) are parked on a horizontal road. A much greater force is needed to push the truck than the car to bring them to the same speed in same time. This is because, for a given time interval, the force is directly proportional to the change in momentum. Here, the change in momentum of truck is quite large than that of the car, therefore, force required for truck is quite large as compared to that required for car.

(ii) An experienced cricketer catches a cricket ball coming in with great speed far more easily than a beginner, who can hurt his hands in the act. One reason is that the cricketer allows a longer time for his hands to stop the ball. He draws in the hands backward in the act of catching the ball (see figure). As the time for catching increases, the force with which the ball hurts his hand decreases. As a result, his hands are not injured. A beginner, on the other hand, keeps his hands fixed and tries to catch the ball almost instantly (i.e. in a very small time interval). Thus, a much greater force is exerted by the ball on his hands and this hurts a lot.

A cricketer catching the ball

Force, $F = \frac{p_2 - p_1}{t}$,this means that for a given change in momentum, the force is inversely proportional to the time interval.

SPOT LIGHT

(iii) If two stones, one light and the other heavy, are dropped from the top of a building, a person on the ground will find it easier to catch the light stone than the heavy stone. This is because the force is directly proportional to the mass of an object.

(iv) Speed/velocity is another important parameter to consider. A bullet fired by a gun can easily pierce human tissue before it stops, resulting in casualty. The same bullet fired with moderate speed will not cause much damage. Thus for a given mass, the greater the speed, the greater is the opposing force needed to stop the body in a certain time.

(v) When an athlete goes for a high jump, he is made to fall on a cushioned bed. This increases the time of falling of the athlete, thereby reducing the force exerted on him, causing no injury.

Check your Concepts 3

1. All of the following, except one, cause the acceleration of an object to double. Which one is it?

(a) All forces acting on the object double.

(b) The net force acting on the object doubles.

(c) Both the net force acting on the object and the mass of the object double.

(d) The mass of the object is reduced by a factor of two.

2. A greater opposing force is needed to stop a heavy body than a light body in the same time, if they are moving with the same speed. Explain.

3. How seat belts help to prevent injuries to the passengers in car in case of an accident?

Numerical Ability 2

1. How much force should be applied to a body of mass 250 g to produce an acceleration of 12 m/s^2 in it?

Solution

Given mass, m = 250 g = $\frac{250}{1000}$ kg = 0.25 kg

Acceleration, a = 12 m/s^2

force, F = ma

= 0.25 × 12 = **3 N**

Decode the problem

Identify the given quantity

Mass ✓

Acceleration ✓

Apply the formula

F = ma

2. A cricket ball of 0.2 kg moving with a speed of 40 m/s is brought to rest by player in 0.05 s. Find the average force applied by the player.

Solution

Given, mass of ball, m = 0.2 kg; initial velocity, u = 40 m/s; final velocity, v = 0; time, t = 0.05 s

Now, initial momentum, p_1 = mu = 0.2 × 40 = 8 kg m/s

Final momentum, p_2 = mv = 0.2 × 0 = 0 kg m/s

Force, $F = \frac{p_2 - p_1}{t} = \frac{0-8}{0.05}$

$= -\frac{800}{5} = $ **–160 N**

Decode the problem

Identify the given quantity

Mass ✓

Initial velocity ✓

Final velocity ✓

Time ✓

Apply the formula

$F = \frac{p_2 - p_1}{t} = \frac{m(v-u)}{t}$

3. Estimate the net force needed to accelerate a 1000 kg car at 5 m/s^2. If same force is applied to a 200 g ball, what will be its acceleration?

Solution

Given, mass of car, m = 1000 kg; acceleration of car, a = 5 m/s^2 ; force needed, F =? Force, F = ma = 1000 × 5 = 5000 N

Now, same force is applied to a 200 g ball i.e., F = 5000 N

mass of ball, m' = 200 g = 200/1000 = 0.2 kg

Thus, acceleration of ball, $a' = \frac{F}{m'} = \frac{5000}{0.2}$ = **2.5 × 10^4 m/s^2**

Decode the problem

Identify the given quantity

Mass ✓

Acceleration ✓

Apply the formula

F = ma

4. A car of mass 800 kg, moving with velocity 10 m/s is brought to rest in 20 seconds by applying brakes. Calculate the force acting on the wheels.

Solution

Given, mass, m = 800 kg; initial velocity, u = 10 m/s;

final velocity, v = 0; time, t = 20 s

$$f = ma = \frac{m(v-u)}{t} = \frac{800\times(0-10)}{20} \qquad (\because\ a = \frac{v-u}{t})$$

$$= \frac{800\times(-10)}{20}$$

= **– 400 N**

Decode the problem

Identify the given quantity

Mass ✓

Initial velocity ✓

Final velocity ✓

Time ✓

Apply the formula

$$F = \frac{p_2 - p_1}{t} = \frac{m(v-u)}{t}$$

5. A constant force acts on an object of mass 5 kg for a duration of 2 s. It increases the object's velocity from 3 m s^{-1} to 7 m s^{-1}. Find the magnitude of the applied force. If the force was applied for a duration of 5 s, what would be the final velocity of the object?

Solution

Given, initial velocity, u = 3 m s^{-1} ; final velocity, v = 7 ms^{-1}

time, t = 2 s ; mass of the object, m = 5 kg ; force, F =?

$$\text{Acceleration, } a = \frac{v-u}{t} = \frac{7-3}{2} = \frac{4}{2} = 2 \text{ m s}^{-2}$$

Force, F = ma = 5 × 2 = 10 N

If same force was applied for a duration of 5 s, the acceleration will remain the same i.e., a = 2 m s^{-2}

Now, from first equation of motion,

v = u + at = 3 + (2 × 5) = 3 + 10 = **13 m s^{-1}**

Decode the problem

Identify the given quantity

Mass ✓

Acceleration ✓

Apply the formula

F = ma

Identify the formula for final velocity

v = u + at

6. A force of 5 N gives a mass m_1 an acceleration of 10 m s^{-2} and a mass m_2, an acceleration of 20 m s^{-2}. What acceleration would it give if both the masses were tied together?

Solution

Given, m_1 =? ; force on m_1 , F = 5 N

acceleration on m_1, a_1 = 10 m s^{-2} ; m_2 =?

force on m_2, F = 5 N ; acceleration on m_2, a_2 = 20 m s^{-2}

$$\text{We know that, } F = ma \text{ or } m = \frac{F}{a}$$

$$\text{Thus, } m_1 = \frac{F}{a_1} = \frac{5}{10} = 0.5 \text{ kg}.$$

$$\text{Similarly, } m_2 = \frac{F}{a_2} = \frac{5}{20} = 0.25 \text{ kg}$$

Now, if both the masses are tied together and same force F is applied on them, then,

$$F = (m_1 + m_2)\,a', \text{ or } \quad a' = \frac{F}{m_1+m_2} = \frac{5}{0.5+0.25} = \frac{5}{0.75} = \mathbf{6.67\ m\ s^{-2}}$$

Decode the problem

Identify the given quantity

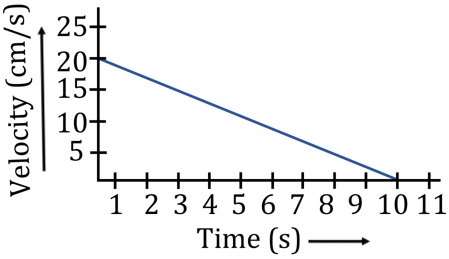
Force ✓

Acceleration ✓

Apply the formula

F = ma

7. The velocity-time graph of a ball of mass 20 g moving along a straight line on a long table is given in figure. How much force does the table exert on the ball to bring it to rest?



Numerical ability 2

Solution

The initial velocity of the ball is 20 cm s^{-1}. Due to the friction force exerted by the table, the velocity of the ball decreases down to zero in 10 s (as per graph).

Initial velocity, u = 20 cm s^{-1} = (20/100) m s^{-1} = 0.2 m s^{-1} ; final velocity, v = 0 ; time, t = 10 s ;

mass of the ball, m = 20 g = (20/1000) kg = 0.02 kg

Since the velocity-time graph is a straight line, it is clear that the ball comes to rest with a constant acceleration.

Acceleration, $a = \dfrac{v-u}{t} = \dfrac{0-0.2}{10} = -0.02 \text{ m s}^{-2}$

Force exerted on the ball, F = ma = 0.02 × (–0.02) = **-4×10^{-4} N**

The negative sign implies that the frictional force exerted by the table is opposite to the direction of motion of the ball.

Decode the problem

Identify the given quantity

Initial velocity ✓

Final velocity ✓

Time ✓

Apply the formula

$a = \dfrac{v-u}{t}$

F = ma

Check your Answers 3

1. (a) Since all the forces acting on the object double, the net force acting on the object also doubles, thus, the acceleration of the object doubles.

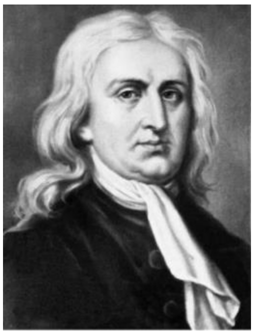
(b) Since the net force acting on the object doubles, thus, the acceleration of the object doubles.

(c) Since both net force acting on the object and the mass of the object double, the acceleration remains the same (F = ma).

(d) The acceleration produced in the object is inversely proportional to the mass ($a \propto 1/m$). Therefore, when the mass of an object is made half the original value, its acceleration gets doubled. Thus, in case of option (c), the acceleration is not the double of its original value.

2. Since the heavy body has greater momentum than the light body at the same speeds, the change in momentum of the heavy body is also greater than that of the lighter body. For a given time interval, the force is directly proportional to the change in momentum. Therefore, the force needed to stop a heavy body is greater than a light body in the same time.

3. In case of an accident or sudden application of brakes, the change in momentum is very large. The seat belts or safety belts worn by the passengers exert force on their body and make the forward motion slower. That is, the time taken by the passengers to fall forward increases, hence the force exerted on them decreases. They may suffer minor injuries or no injuries at all.



Sir Isaac Newton, English physicist and mathematician (1642-1727)

9. Newton's third law of motion

The second law relates the external force on a body to its acceleration. What is the origin of the external force on the body? What agency provides the external force? The simple answer is that **the external force on a body always arises due to some other body**.

Forces always exist in pairs

When two objects interact, two forces will always be involved. One force is the **action force** and the other is the **reaction force**.

- According to Newton's third law, **'whenever one body exerts a force on a second body, the second body exerts an oppositely directed force of equal magnitude on the first body'**.

  In other words, **'to every action, there is always an equal and opposite reaction'**. Consider a pair of bodies A and B. According to Newton's third law, $\mathbf{F_{AB} = -F_{BA}}$, where, F_{AB} = force on A due to B, and F_{BA}= force on B due to A.

- An important aspect of Newton's third law is that the action and reaction forces act on different bodies. This means that action-reaction pairs will never be added together i.e., they can not cancel out each other.

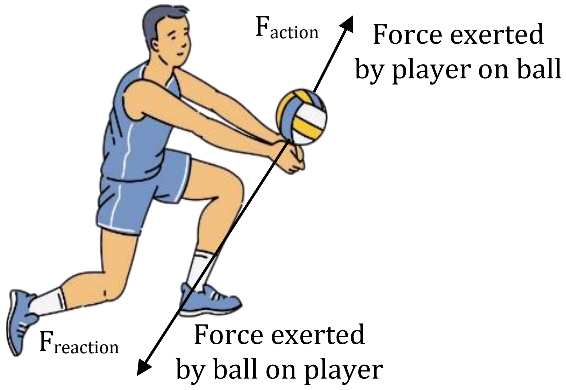
Building Concepts 5

If the action force is equal in magnitude to the reaction force, how can there ever be an acceleration? Explain using an example.

Explanation

Though the action-reaction pair are equal in magnitude and opposite in direction but the reaction force always acts on a different object than the action force. Thus, these forces do not cancel out each other. Hence, there can be an acceleration in an object.

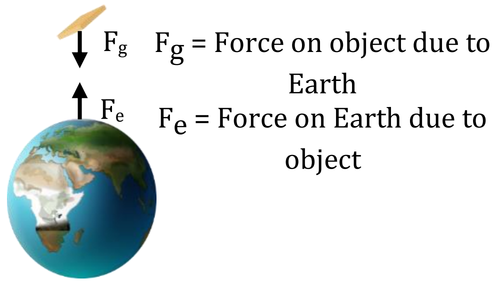
Example : Let us consider a volleyball player bumping the ball (see figure). At the instant when both the ball and the player's arms are in contact, the action force is the upward force that the player exerts on the ball. The reaction force is the downward force that the ball exerts on the player's arms. During the collision, the ball accelerates upward and the player's arms accelerate downward. We hardly notice the acceleration of player's arms since his mass is quite large as compared to the ball, and the effect of the force on his motion is negligible.



Building concepts 5

Examples/applications of Newton's third law

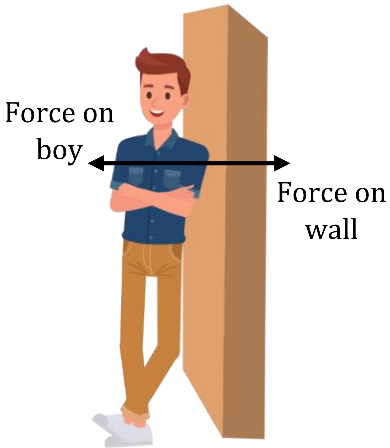
1. Newton's third law is also applicable to non-contact forces. For example, the Earth pulls an object downwards due to gravity (see figure). The object's force also exerts the same force on the Earth but in upward direction. But, we hardly see the effect of the object on the Earth because the Earth is very massive and the effect of a small force on its state of rest or motion is negligible. That is, the acceleration of Earth is negligible due to its huge mass.



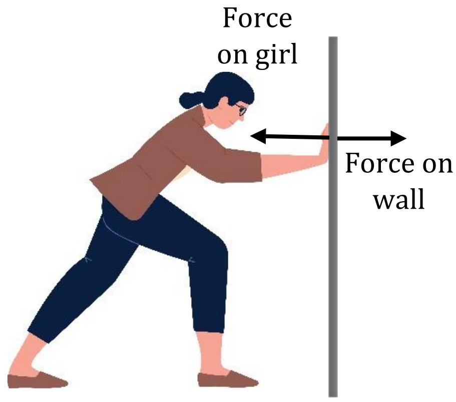
Newton's third law is also applicable to non-contact forces

2. If boy exerts a small force on the wall, the wall will exert a small force on boy [see figure (a)]. When girl pushes hard against the wall, it pushes back just as hard [see figure (b)].
3. When a sailor jumps out of a rowing boat. As the sailor jumps forward, the force on the boat moves it backwards [see figure (c)].
4. When a gun is fired, it exerts a forward force on the bullet. The bullet exerts an equal and opposite reaction force on the gun. This results in the recoil of the gun [see figure (d)]. Since the gun has a much greater mass than the bullet, the acceleration of the gun is much less than the acceleration of the bullet.

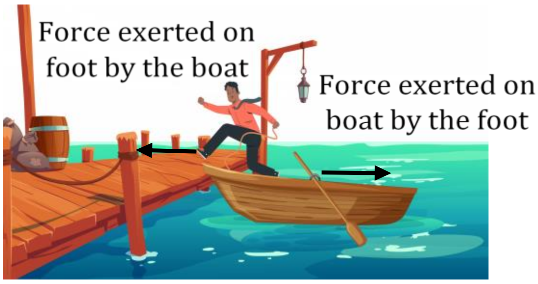
5. Suppose you are standing at rest and intend to start walking (or running) on a road [see figure (e)]. While walking, you push the road backwards. Thus, according to Newton's third law, the road exerts an equal and opposite reaction force on your feet to make you move forward.



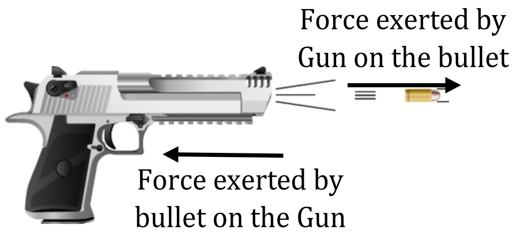
(a)



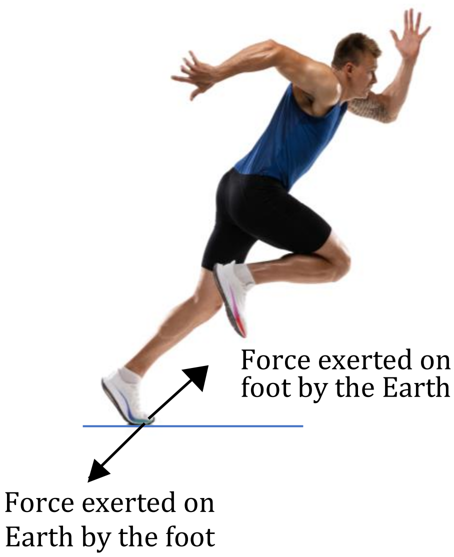
(b)



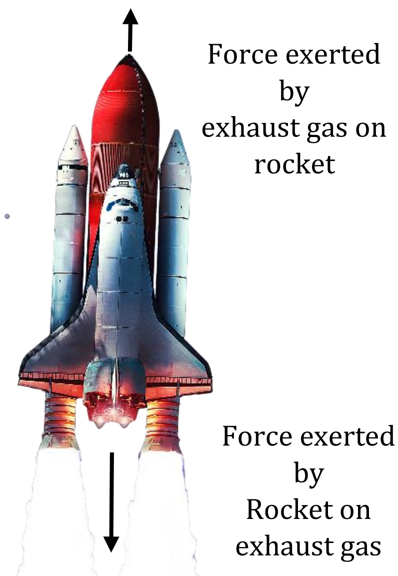
(c)



(d)



(e)



(f)Action-reaction pairs in rocket propulsion

Understanding Newton's third law

6. **Rocket Propulsion :** In a rocket engine, the highly combustible fuel burns at a tremendous rate [see figure (f)]. The rocket exerts a downward (or backward) force on the exhaust gas and thus, according to Newton's third law, the exhaust gas exerts an upward (or forward) force on the rocket, these forces are equal in magnitude. It is the reaction force of the exhaust gas that accelerates the rocket forward. That is why a rocket can accelerate even in outer space.

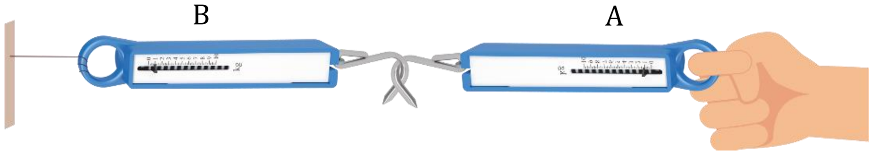
Check your Concepts 4

1. A high-speed car and an innocent insect have a head-on collision. The force of impact splatters the poor insect over the windshield. Is the corresponding force that the insect exerts against the windshield greater, less, or the same? Is the retardation of the car greater than, less than, or the same as that of the insect?

Active Physics 4

Take two spring balances connected together as shown in figure The fixed end of balance B is attached with a rigid support, like a wall. Apply a force through the free end of spring balance A. You will observe that both the spring balances show the same readings on their scales.

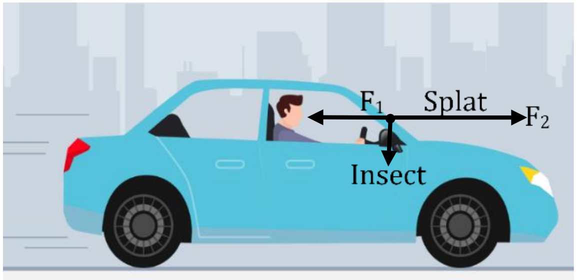
Conclusion : The force exerted by spring balance A on balance B is equal but opposite in direction to the force exerted by the balance B on balance A. The force which balance A exerts on balance B is called the 'action' and the force of balance B on balance A is called the 'reaction'.



Active physics 4

Check your Answers 4

1. The magnitudes of both forces are the same, they constitute an action-reaction pair that makes up the interaction between the car and the insect (see figure). The accelerations, however, are very different because their masses are different. The insect undergoes an enormous and lethal retardation, while the car undergoes a negligible retardation, so that it cannot be noticed by its driver or passengers.



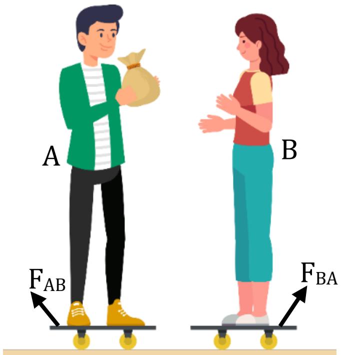
F_1 = Force on car by the insect, F_2 = Force on insect by the car

Check your answers 4 (1)

Active Physics 5

1. Let you and your classmate are standing on two separate carts (see figure). Take a bag full of sand or some other heavy object. Now, play a game of catch with the bag. You and your classmate will receive an instantaneous reaction force as a result of throwing the sand bag (action force). When you throw the bag in forward direction, cart along with you will move in backward direction. Similar thing will happen to your classmate. As a result both of you will move away from each other.

2. This observation is in agreement with Newton's third law. We know that action and reaction forces are equal in magnitude but opposite in direction i.e., $F_{BA} = -F_{AB}$. But the accelerations of you and your classmate will not be the same as acceleration is inversely proportional to the mass for a given force. This means, if you are heavier than your classmate, your acceleration will be less and vice-versa.



Active physics 5

9. Conservation of momentum

The second and the third laws of motion lead to an important result : **the law of conservation of momentum**. Let us understand it by taking an example : When a bullet is fired from a gun, according to Newton's third law, if the force on the bullet by the gun is F, then the force on the gun by the bullet is – F. The two forces act for a common interval of time t.

According to Newton's second law, force F can be written as,

$F = \dfrac{\Delta p}{t}$ where, Δp is the change in momentum of the object.

or $\quad \Delta p = F \times t$

This means, $F \times t$ is the change in momentum of the bullet and $-F \times t$ is the change in momentum of the gun. Since initially, both are at rest, the change in momentum equals the final momentum for each. Let p_b be the momentum of the bullet after firing and p_g is the recoil momentum of the gun, then,

$p_b = F \times t \quad ... (1) \quad$ and $\quad p_g = -F \times t \quad ... (2)$

> Momentum is conserved for all types of collisions that take place in real world whenever external forces don't interfere.
>
> **SPOT LIGHT**

Adding eq. (1) and eq. (2), we get, $p_b + p_g = 0$. That is, the final momentum of the system (bullet plus gun) is zero. But, initial momentum of the system is also zero. This means initial momentum is equal to the final momentum i.e., **total momentum is conserved**.

Thus, in an isolated system (a system with no external force), mutual forces between pairs of particles in the system can cause momentum change in individual particles, but since the mutual forces for each pair are equal and opposite, the momentum changes cancel in pairs and the total momentum remains unchanged. This fact is known as the **law of conservation of momentum**.

☞ According to the law of conservation of momentum, **'when the net external force on a system of objects is zero, the total momentum of the system remains constant'**.

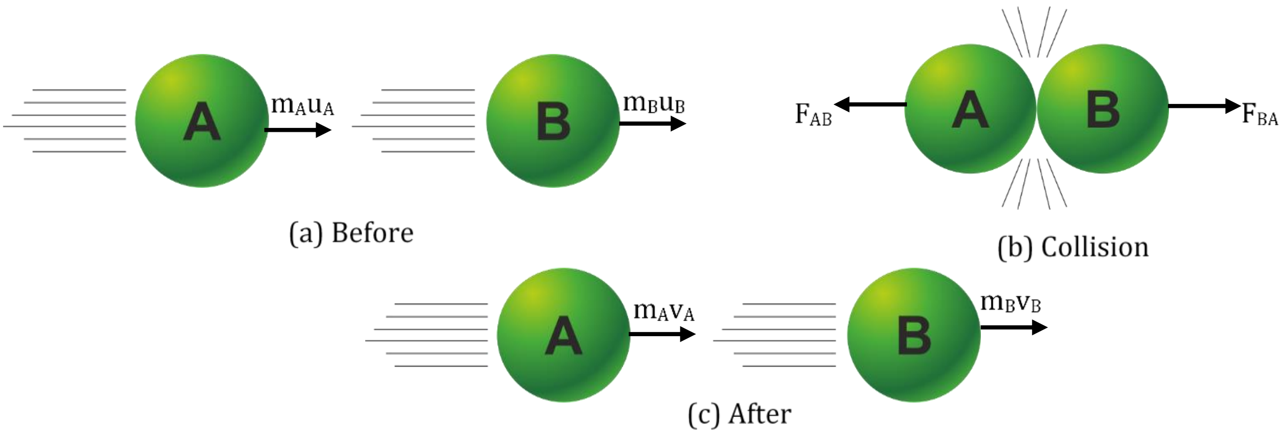
The total momentum of an isolated system of objects remains constant. This is an another statement of the law of conservation of momentum.

SPOT LIGHT

The term 'collision' is used to represent the event of two particles coming together for a short time and thereby producing 'impulsive forces' on each other. These forces are assumed to be much greater than any external forces present because they act for a very short time interval.

Law of conservation of momentum : proof

Let us consider two balls A and B having masses m_A and m_B, travelling in the same direction along a straight line with different velocities u_A and u_B, respectively [see figure (a)]. No other external unbalanced forces are acting on them. Let $u_A > u_B$ and the two balls collide with each other as shown in figure (b). During collision which lasts for a very short time t, the ball A exerts a force F_{BA} on ball B, and the ball B exerts a force F_{AB} on ball A. Suppose v_A and v_B are the velocities of the two balls A and B after the collision, respectively [see figure(c)].



Proving law of conservation of momentum

Initial momentum of ball A = m_Au_A, and final momentum of ball A = m_Av_A Force on A due to B, F_{AB} = rate of change of momentum of ball A

or $F_{AB} = \frac{m_Av_A - m_Au_A}{t} = \frac{m_A(v_A - u_A)}{t}$ ...(1)

Initial momentum of ball B = m_Bu_B,

Final momentum of ball B = m_Bv_B

Force on B due to A, F_{BA} = rate of change of momentum of ball B

or $F_{BA} = \frac{m_Bv_B - m_Bu_B}{t} = \frac{m_B(v_B - u_B)}{t}$ ...(2)

According to Newton's third law of motion, the force F_{BA} exerted by ball A on ball B (action) and the force F_{AB} exerted by the ball B on ball A (reaction) must be equal and opposite to each other. That is, $F_{BA} = -F_{AB}$

$$\frac{m_B(v_B - u_B)}{t} = -\frac{m_A(v_A - u_A)}{t}$$

or $m_B(v_B - u_B) = -m_A(v_A - u_A)$

or $m_Bv_B - m_Bu_B = -m_Av_A + m_Au_A$

or $\mathbf{m_Bv_B + m_Av_A = m_Bu_B + m_Au_A}$ ...(3)

Since, $(m_Bv_B + m_Av_A)$ represent the total final momentum and $(m_Bu_B + m_Au_A)$ represent the total initial momentum, from eq. (3), we can conclude that

Total final momentum = Total initial momentum

This is **law of conservation of momentum**.

Check your Concepts 5

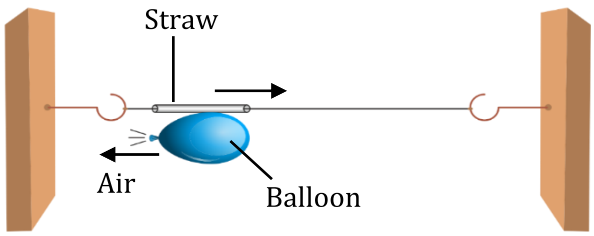
1. Newton's second law states that, if no net force is exerted on a system, no acceleration occurs. Does it follow that no change in momentum occurs?
2. In boxing, if a boxer increases the duration of impact of his punch on the opponent, how does the force of impact change?
3. Explain, the recoiling of the gun, using law of conservation of momentum.

Active Physics 6

1. Take a big rubber balloon and inflate it fully. Tie its neck using a thread. Also using adhesive tape, fix a straw on the surface of this balloon. Pass a thread through the straw and hold one end of the thread in your hand or fix it on the wall (see figure). Fix the other end of the thread on a wall at some distance. Now remove the thread tied on the neck of the balloon. Let the air escape from the mouth of the balloon.

Publishing\PNCF\2024-25\LIVE Module\SET-1\NCERT\9th\Physics\2_Force and Laws of Motion

2. You will observe that the air escapes from the balloon in backward (leftward) direction while the balloon along with the attached straw moves in forward (rightward) direction. This is in agreement with Newton's third law or the law of conservation of momentum.



Active physics 6

Numerical Ability 3

1. A bullet of mass 20 g is horizontally fired with a velocity 150 m s^{-1} from a pistol of mass 2 kg. What is the recoil velocity of the pistol?

Solution

Given, mass of bullet, m_1 = 20 g = 0.02 kg ; mass of pistol, m_2 = 2 kg; initial velocity of bullet, u_1 = 0 ; initial velocity of pistol, u_2 = 0.

Let the direction of bullet is taken left to right (see figure).

Now, the final velocity of the bullet, v_1 = + 150 m s^{-1}

(by sign convention, left to right is taken positive).

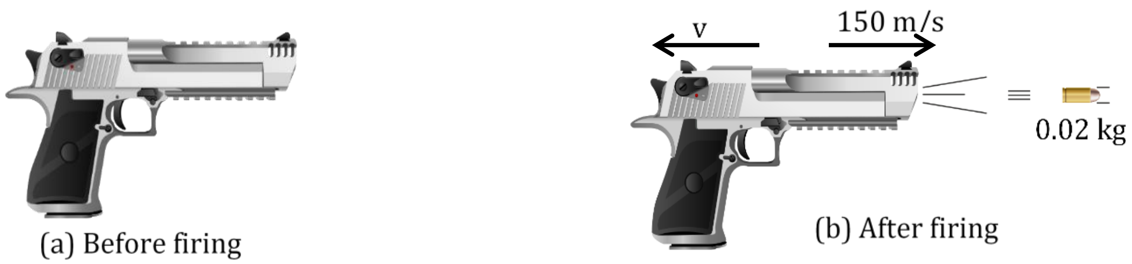
Decode the problem

Identify the concept

The given problem is based on conservation of momentum.

Apply the formula

$m_B v_B + m_A v_A = m_B u_B + m_A u_A$



Numerical Ability 3 (1)

Let 'v' be the recoil velocity of the pistol.

Initial momentum of the pistol-bullet system,

p_i = initial momentum of the pistol + initial momentum of the bullet

$= 2 \times 0 + 0.02 \times 0 = 0$ kg m s^{-1}

Final momentum of the pistol-bullet system,

p_f = final momentum of the pistol + final momentum of the bullet

$= 2 \times v + 0.02 \times 150 = (2v + 3)$ kg m s^{-1}

According to the law of conservation of momentum,

final momentum of the system = initial momentum of the system

$\therefore$ 2v + 3 = 0 or 2v = –3 or v = –3/2 = **–1.5 m/s**

(Negative sign indicates that the direction in which the pistol would recoil is opposite to that of bullet, that is, right to left).

2. A boy of mass 40 kg jumps with a horizontal velocity of 5 m s^{-1} onto a stationary cart with frictionless wheels. The mass of the cart is 3 kg. What is his velocity as the cart starts moving? Assume that there is no external unbalanced force working in the horizontal direction.

Solution

Given, mass of boy, m_1 = 40 kg ; initial velocity of boy, u_1 = + 5 m/s ; mass of cart, m_2 = 3 kg ; initial velocity of cart, u_2 = 0 [see figure (a)].

Initial momentum of the boy-cart system, p_i = Initial momentum of the boy + Initial momentum of the cart

or $p_i = m_1u_1 + m_2u_2 = (40) \times (+5) + (3) \times (0)$

= 200 kg m/s

Decode the problem

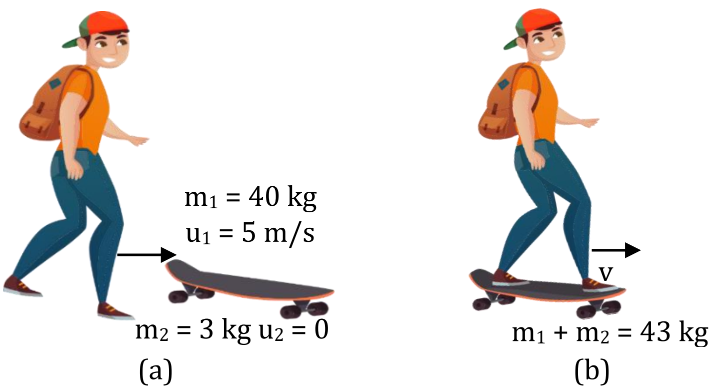
Identify the concept

The given problem is based on conservation of momentum.

Apply the formula

$m_B v_B + m_A v_A = m_B u_B + m_A u_A$

Let 'v' be the common velocity of the boy-cart system when the cart starts moving along with the boy [see figure (b)].



Numerical Ability 3 (2)

Final momentum of the boy-cart system,

p_f = Final momentum of the boy + Final momentum of the cart or $p_f = m_1v + m_2v$

$= (m_1 + m_2)\,v = (40 + 3)\,v = 43\,v$ kg m/s

According to the law of conservation of momentum,

final momentum of the system = initial momentum of the system

or $43\ v = 200$, or

$v = 200/43 = \mathbf{+\ 4.65\ m/s}$

The boy on cart will move with a velocity of **4.65 m/s** in the direction in which the boy jumped.

3. Consider a large fish that swims towards and swallows a small fish at rest (see figure). If the large fish has a mass of 5 kg and swims at 1 m/s toward the small fish having mass of 1 kg, what is the velocity of the large fish immediately after the lunch? Neglect the effects of water resistance.

Numerical Ability 3 (3)

Solution

Given, mass of large fish, $m_1 = 5$ kg ; initial velocity of large fish, $u_1 = 1$ m/s ; mass of small fish, $m_2 = 1$ kg ; initial velocity of small fish, $u_2 = 0$.

Let 'v' be the velocity of the large fish-small fish system.

Applying law of conservation of momentum, we get, total momentum after lunch = total momentum before lunch

or $m_1 v + m_2 v = m_1u_1 + m_2u_2$

or $(m_1 + m_2)\ v = m_1u_1 + m_2u_2$

or $(5 + 1)\ v = (5) \times (+1) + (1) \times (0)$

or $6\ v = 5$,

or $v = (5/6)$ m/s

or $v = \mathbf{+\ 0.833\ m/s}$

The large fish will move with a velocity of **0.833 m/s** in the direction in which it was moving initially.

Decode the problem

Identify the concept

The given problem is based on conservation of momentum.

Apply the formula

$m_B v_B + m_A v_A = m_B u_B + m_A u_A$

4. Two ice hockey players of opposite teams, while trying to hit a hockey ball on the ice collide and immediately become entangled. One has a mass of 60 kg and was moving with a velocity 5.0 m/s while the other has a mass of 55 kg and was moving faster with a velocity 6.0 m/s towards the first player. In which direction and with what velocity will they move after they become entangled? Assume that the frictional force acting between the feet of the two players and ice is negligible.

Solution

Let the first player be moving towards right and the second player moving towards left [see figure (a)].

mass of 1st player, m_1 = 60 kg ; initial velocity of 1st player, u_1 = + 5 m/s ;

mass of 2nd player, m_2 = 55 kg ; initial velocity of 2nd player, u_2 = – 6 m/s.

Decode the problem

Identify the concept

The given problem is based on conservation of momentum.

Apply the formula

$m_B v_B + m_A v_A = m_B u_B + m_A$

Let the common velocity of both the players after collision be 'v' [see figure(b)].

According to the law of conservation of momentum,

final momentum of the system = initial momentum of the system

or $m_1 v + m_2 v = m_1u_1 + m_2u_2$

or $(m_1 + m_2) v = m_1u_1 + m_2u_2$

or $(60 + 55) v = 60 \times (+ 5) + (55) \times (- 6)$

or $115 v = 300 - 330$,

or $115 v = -30$

or $v = -30/115 =$ **– 0.26 m/s**

Negative sign shows that the two entangled players will move towards left, i.e., in the direction the second player was moving before the collision.

(a) Before collision (b) After collision

Numerical Ability 3 (4)

Check your Answers 5

1. Yes, because no acceleration means that no change occurs in velocity. This means there is no change in momentum (= mass × velocity).
2. The force of impact will be less as compared to the initial force.
3. Initially, the gun as well as bullet is at rest i.e., initial momentum is zero. Let the masses of bullet and gun are m_b and m_g respectively. When the bullet is fired from the gun, it acquires a velocity v_b.

   Let the velocity acquired by the gun be v_g. Now, the final momentum will be $(m_b v_b + m_g v_g)$. Since, no external forces are involved in the process, thus, momentum is conserved.

   Therefore,

   final momentum = initial momentum

   or $m_b v_b + m_g v_g = 0$

   or $m_g v_g = - m_b v_b$ or $v_g = -\frac{m_b v_b}{m_g}$ ...(1)

   The negative sign in eq. (1) shows that the velocity of the gun is in opposite direction to the velocity of bullet i.e., the gun recoils when it is fired.

   $m_g >> m_b$, so $v_g << v_b$ i.e. the recoil velocity of the gun is much smaller than the forward velocity of the bullet.

Active Physics 7

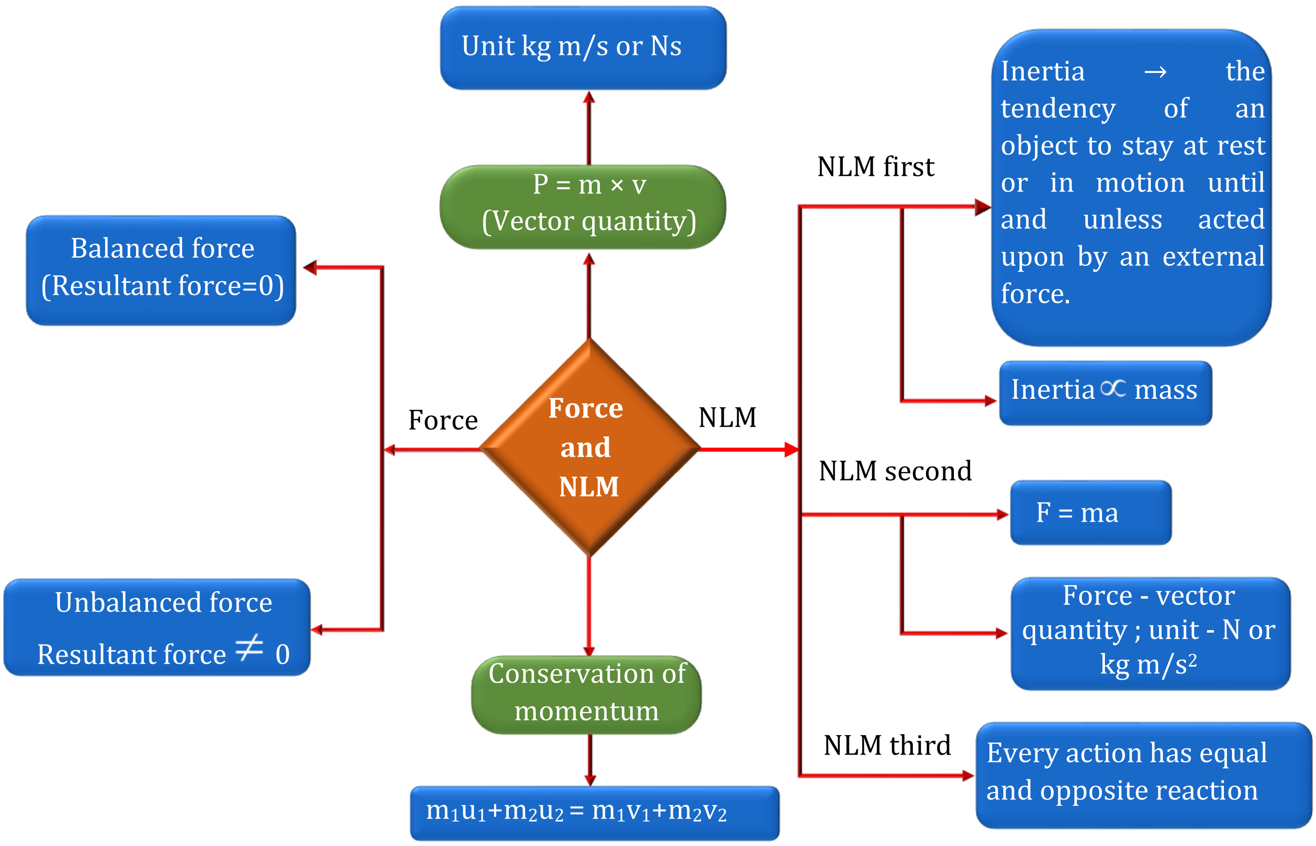
1. Take a test tube of good quality glass material and put a small amount of water in it (see figure). Place a stop cork at its mouth. Now suspend the test tube horizontally by two strings or wires.
2. Heat the test tube with a burner until water vaporises, and due to the pressure of water vapours (steam), the cork blows out. You will observe that the test tube recoils in the direction opposite to the direction of the cork. This observation is in agreement with the law of conservation of momentum. You will observe that the velocity of cork (v_c) is much larger than the velocity (v_t) of the test tube. This is because the mass of cork is quite smaller than the mass of the test tube (for a given value of momentum, velocity is inversely proportional to mass).

Publishing\PNCF\2024-25\LIVE Module\SET-1\NCERT\9th\Physics\2_Force and Laws of Motion

(a) (b)

Active physics 7

Concept Map


Unit kg m/s or Ns
P = m × v
(Vector quantity)
Balanced force
(Resultant force=0)
Force
Force
and
NLM
NLM
Unbalanced force
Resultant force ≠ 0
Conservation of
momentum
$m_1u_1+m_2u_2 = m_1v_1+m_2v_2$
NLM first
Inertia → the tendency of an object to stay at rest or in motion until and unless acted upon by an external force.
Inertia ∝ mass
NLM second
F = ma
Force - vector quantity ; unit - N or kg m/s²
NLM third
Every action has equal and opposite reaction

Some Basic Terms

1. **Muscular force** – The contact force applied to an object with the help of muscles is known as muscular force.
2. **Tension** – It is define as the force transmitted through a rope, string or wire when pulled by forces acting from opposite sides.
3. **Friction** – It is the force generated by two surfaces that contact slide against each other.
4. **Electrostatic force** – The force exerted by a charged body on another charged or uncharged body by virtue of its charged is known as electrostatic force.
5. **Magnetic force** – Magnetic force can be defined as the attractive or repulsive force that is exerted between the poles of a magnet and electrically charged moving particles.
6. **Gravitational force** – The force of attraction that arises between two bodies having definite masses.
7. **Distort**– Distort is to change the physical shape of something.
8. **Stationary** – Stationary means not moving, not appearing to move, stable of fixed.
9. **Frictionless** – In physics, frictionless means that there is no resistance between a surface or substance and something that is moving along or through it. In other words, without any friction.
10. **Mechanics** – It is a branch of science which deals with the motion of an object and the forces acting on it.
11. **Jerk** – A sudden, quick, sharp movement.
12. **Tumbler** – A tall glass for drinking out of with straight side and no handle.
13. **Collision** – A brief event in which two or more bodies come together. For example : Two billiards ball, a golf club and a ball, a hammer and a nail head, or a falling object and a floor
14. **Head on collision** – A crash of two objects that are moving directly towards each other.
15. **Inflates** – It means to add air or gas to something such as a tire or a balloon and make it larger.
16. **External force** – It is a force that act on a structure from the outside. For example : wind pushing on a tree is an external force.
17. **Internal force** – Any force that acts on a structure from with in known as a internal force. Internal forces not change the mechanical energy of an object.
18. **Resultant force** – The resultant force is described as the total amount of force acting on the object or body along with a direction of the body.
19. **Recoil** – Recoil is a movement backwards, usually from some force or impact.
    For example : The recoil of a gun is backward movement caused by momentum.

SOLVED EXAMPLES

1. An object 'A' of mass 2 kg is moving with a uniform velocity of 30 m/s. If an another object 'B' of 15 kg mass moves with the same momentum as of object A, find the velocity of object B.

Solution

Here, $m_A = 2$ kg, $v_A = 30$ m/s,

$m_B = 15$ kg, $v_B = ?$

momentum of object A,

$p_A = m_A v_A = 2 \times 30 = 60$ kg m/s

Given that $p_A = p_B$, therefore $p_B = 60$ kg m/s

$\because\ p_B = m_B v_B = 15 \times v_B$

$\therefore\ 60 = 15 \times v_B$

or $v_B = \dfrac{60}{15} = \mathbf{4\ m/s}$

2. Calculate the force required to provide a velocity of 30 m s^{-1} to a car in 10 seconds. The mass of the car is 1500 kg.

Solution

Here, u = 0 m s^{-1}; v = 30 ms^{-1}; t = 10 s;

a =?

Using v = u + at, we have

30 = 0 + a (10)

$\therefore$ a = 3 m s^{-2}

Now, F = ma = 1500 × 3

or F = **4500 N**

3. A cricket ball of mass 70 g moving with a velocity of 0.5 m s^{-1} is stopped by player in 0.5 s. What is the force applied by player to stop the ball?

Solution

Here, m = 70 g = $\dfrac{70}{1000}$ = 0.07 kg ;

u = 0.5 m s^{-1} ; v = 0 ; t = 0.5 s

$F = \dfrac{m(v-u)}{t}$ or $F = \dfrac{0.07(0-0.5)}{0.5}$

or F = **– 0.07 N**

4. A bullet of mass 10 g is fired from a rifle. The bullet takes 0.003 s to move through its barrel and leaves with a velocity of 300 m s^{-1}. What is the force exerted on the bullet by the rifle?

Solution

Here, m = 10 g = 0.01 kg ; u = 0 ;

v = 300 m s^{-1}, t = 0.003 s, F =?

$$F = \frac{m(v-u)}{t}$$

$$\text{or } F = \frac{0.01 \times (300-0)}{0.003}$$

or F = **1000 N**

5. How long should a force of 100 N act on a body of 20 kg, so that it acquires a velocity of 100 m s^{-1}?

Solution

Here, u = 0 ; v =100 m s^{-1}, m = 20 kg ;

F = 100 N ;t =?

We know $F = ma = \dfrac{m(v-u)}{t}$

$$\text{or } t = \frac{m(v-u)}{F} = \frac{20(100-0)}{100}$$

= **20 s**

6. A 1,000 kg vehicle moving with a speed of 20 m s^{-1} is brought to rest in a distance of 50 m. Find (i) the acceleration of vehicle (ii) the unbalanced force acting on the vehicle The actual force applied by the brakes may be slightly less than that calculated in (ii). Why? Give reason.

Solution

(i) Here u = 20 m s^{-1}; v = 0; s = 50 m; a =?

Using $v^2 - u^2 = 2as$, we have

$a = \dfrac{v^2 - u^2}{2s} = \dfrac{0-(20)^2}{2(50)} = \mathbf{-4\ ms^{-2}}$

(ii) F = ma = 1,000 × (–4) = –4,000 N

Due to force of friction which assists in slowing down the vehicle, the actual force applied by brakes may be slightly less than calculated one.

7. Which would require greater force: accelerating a 10 g mass at 5 m s^{-2}, or 20 g mass at 2 m s^{-2}?

Solution

In first case

$m_1 = 10\text{ g} = \frac{10}{1,000}$ kg = 0.01 kg;

Now $a_1 = 5\text{ ms}^{-2}$; F_1 =?

$F_1 = m_1 a_1 = 0.01 \times 5$

$F_1 = 0.05$ N

In second case, $m_2 = 20\text{ g} = \frac{20}{1,000}$ kg

= 0.02 kg

$a_2 = 2\text{ m s}^{-2}$; F_2 = ?

Now $F_2 = m_2 a_2 = 0.02 \times 2$

or $F_2 = 0.04$ N

We find that $F_1 > F_2$, hence more force is required to accelerate a 10 g mass at 5 m s^{-2}, than a accelerating 20 g mass at 2 m s^{-2}.

8. A car of mass 1,000 kg moving with a velocity of 54 km h^{-1} collides with a tree and comes to a stop in 5 s. What will be the force exerted by car on the tree?

Solution

Here m = 1000 kg

$u = 54 \times \frac{5}{18}\text{ ms}^{-1} = 15\text{ ms}^{-1}$

$v = 0$; $t = 5$ s

$\therefore\ F = \frac{m(v-u)}{t}$

or $F = \frac{1000\times(0-15)}{5}$

or F = 200 × (–15)

or F = –3000 N= force exerted by tree on car

$\therefore$ Force exerted by car on tree = 3000 N

9. A bullet of mass 100 g is fired from a gun of mass 20 kg with a velocity of 100 ms^{-1}. Calculate the recoil velocity of the gun.

Solution

Mass of bullet,

m = 100 g

$= \frac{100}{1000}\text{ kg} = \frac{1}{10}\text{ kg}$

Velocity of bullet,

v = 100 ms^{-1}

Mass of gun, M = 20 kg

Let, recoil velocity of gun = V

Before firing, the system (gun + bullet) is at rest, therefore, initial momentum of the system = 0

Final momentum of the system

= momentum of bullet + momentum of gun

= mv + MV

$= \frac{1}{10} \times 100 + 20\text{ V}$

= 10 + 20 V

Applying law of conservation of momentum,

Final momentum = Initial momentum

i.e. 10 + 20 V = 0

or 20 V = – 10

or $V = -\frac{10}{20}$

= – 0.5 ms^{-1}.

Negative sign shows that the direction of recoil velocity of the gun is opposite to the direction of the velocity of the bullet.

10. Two small glass spheres of masses 10 g and 20 g are moving in a straight line in the same direction with velocities of 3 ms^{-1} and 2 ms^{-1} respectively. They collide with each other and after collision, glass sphere of mass 10 g moves with a velocity of 2.5 ms^{-1}. Find the velocity of the second glass sphere after collision.

Solution

Here, $m_1 = 10\text{ g} = \frac{10}{1000}\text{ kg} = 0.01\text{ kg}$

$m_2 = 20\text{ g} = 0.02\text{ kg}$

$u_1 = 3\text{ ms}^{-1}\text{ ; } u_2 = 2\text{ ms}^{-1}$

$v_1 = 2.5\text{ ms}^{-1}\text{ ; } v_2 = ?$

Total momentum of both the spheres before collision = $m_1u_1 + m_2u_2$

$= 0.01 \times 3 + 0.02 \times 2$

$= 0.07\text{ kg ms}^{-1}$

Total momentum of both the spheres after collision = $m_1 v_1 + m_2 v_2$

$= 0.01 \times 2.5 + 0.02\, v_2 = 0.025 + 0.02\, v_2$

Now, according to the law of conservation of momentum,

Total momentum after collision = Total momentum before collision

$\therefore\ 0.025 + 0.02\, v_2 = 0.07$

or $0.02\, v_2 = 0.07 - 0.025 = 0.045$

or $v_2 = \frac{0.045}{0.02}$ **= 2.25 $\mathbf{ms^{-1}}$**